EX.No.-2.2 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

Date : 14/10/26

AIM:

To create a 3D model from the given isometric drawing using Fusion 360 tool.

REQUIREMENTS:

Autodesk Fusion 360
Windows 10
16 GB of RAM (integrated graphics recommend 1 GB or more)
2.5 Mbps or faster download; 500 Kbps or faster upload

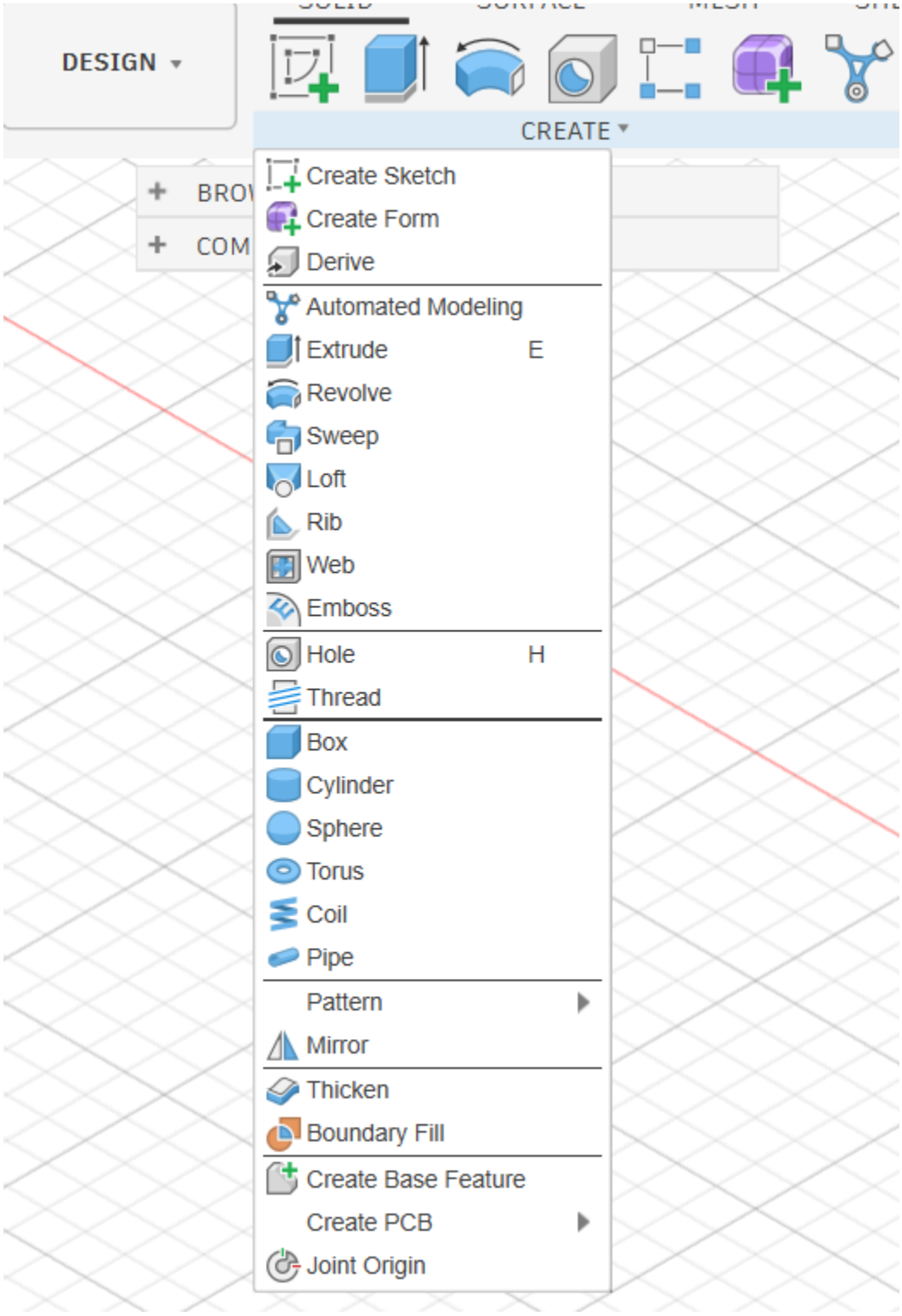
STEP-2: Select the Sketch Plane

Open the Create Sketch command and choose the TOP (XY) Plane to begin drawing the model.

STEP-3: Create a Sketch

Choose Sketch > Rectangle > Center Rectangle and position the cursor at the origin of the sketch plane. Click to start drawing the rectangle. Press the Tab key

and specify the width as 56 mm and height as 18 mm.



STEP-4: Extrude Sketch in XY Plane

Choose the Extrude command from the toolbar and select the completed sketch profile. Specify the extrusion distance as 8 mm and press ENTER to confirm the operation.

STEP-5: Start the sketch command

Select Create Sketch from the toolbar to begin drawing the next profile.

STEP-6: Select the Sketch Plane

Choose the TOP SURFACE of the 3D block as the plane for creating the next sketch.

STEP-7: Create a RECTANGLE

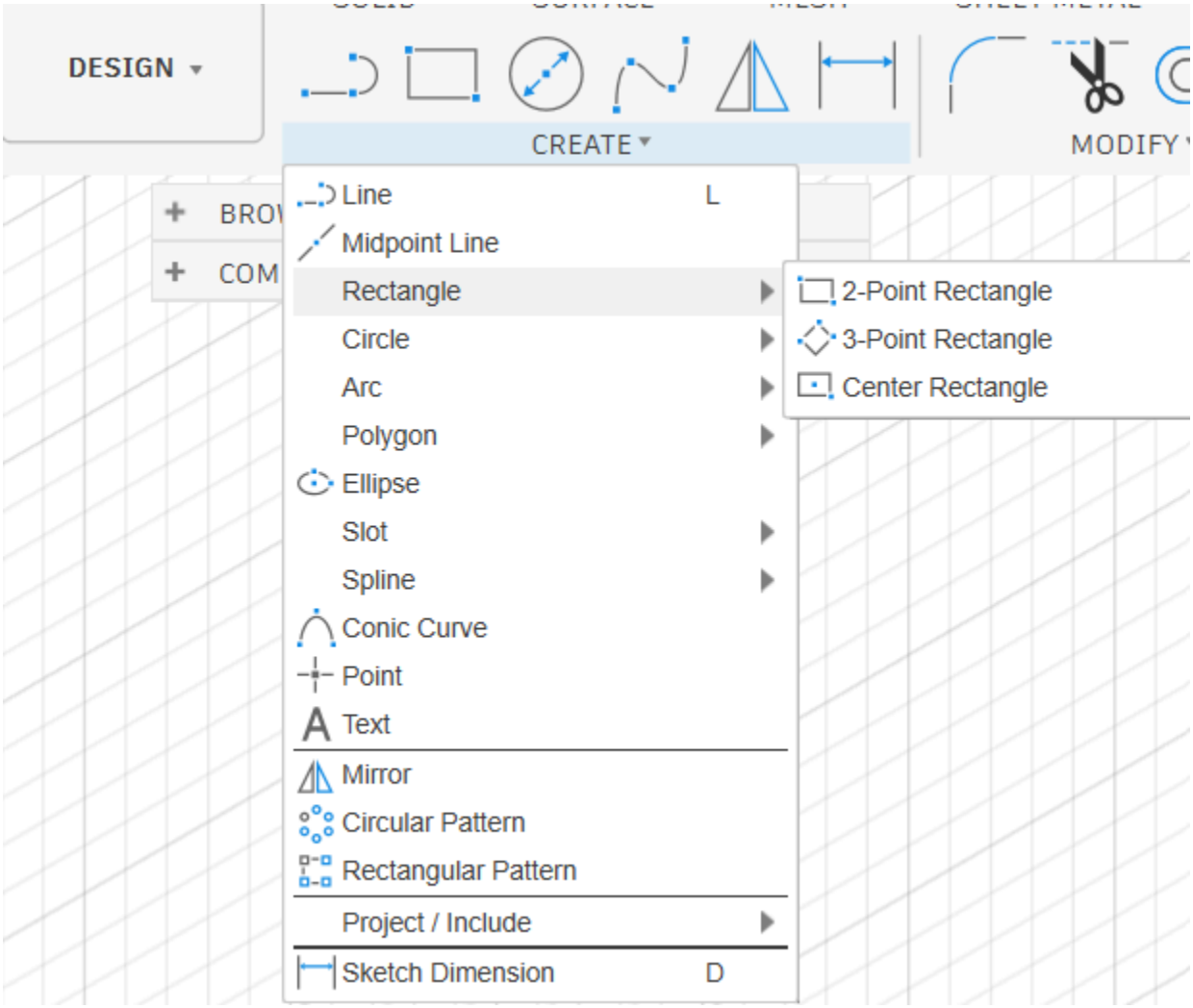
Select Sketch > Rectangle > Center Rectangle and move the cursor to the origin of the sketch. Click to begin drawing and press the Tab key to enter the width as 40 mm and height as 10 mm. Confirm the dimensions and click Finish Sketch.

STEP-8: Extrude RECTANGLE

Activate the Extrude command and select the rectangular sketch profile. Enter the extrusion distance as 32 mm and press ENTER to complete the operation.

STEP-9: Create a TRIANGLE

Select Sketch > Triangle > Center Triangle. Position the cursor at the origin and click to start creating the triangle.



STEP-10: Extrude TRIANGLE

Select the Extrude command and choose the triangular sketch profile. Set the extrusion distance to 6 mm in both directions and press ENTER to apply the feature.

STEP-11: Create a PENTAGON

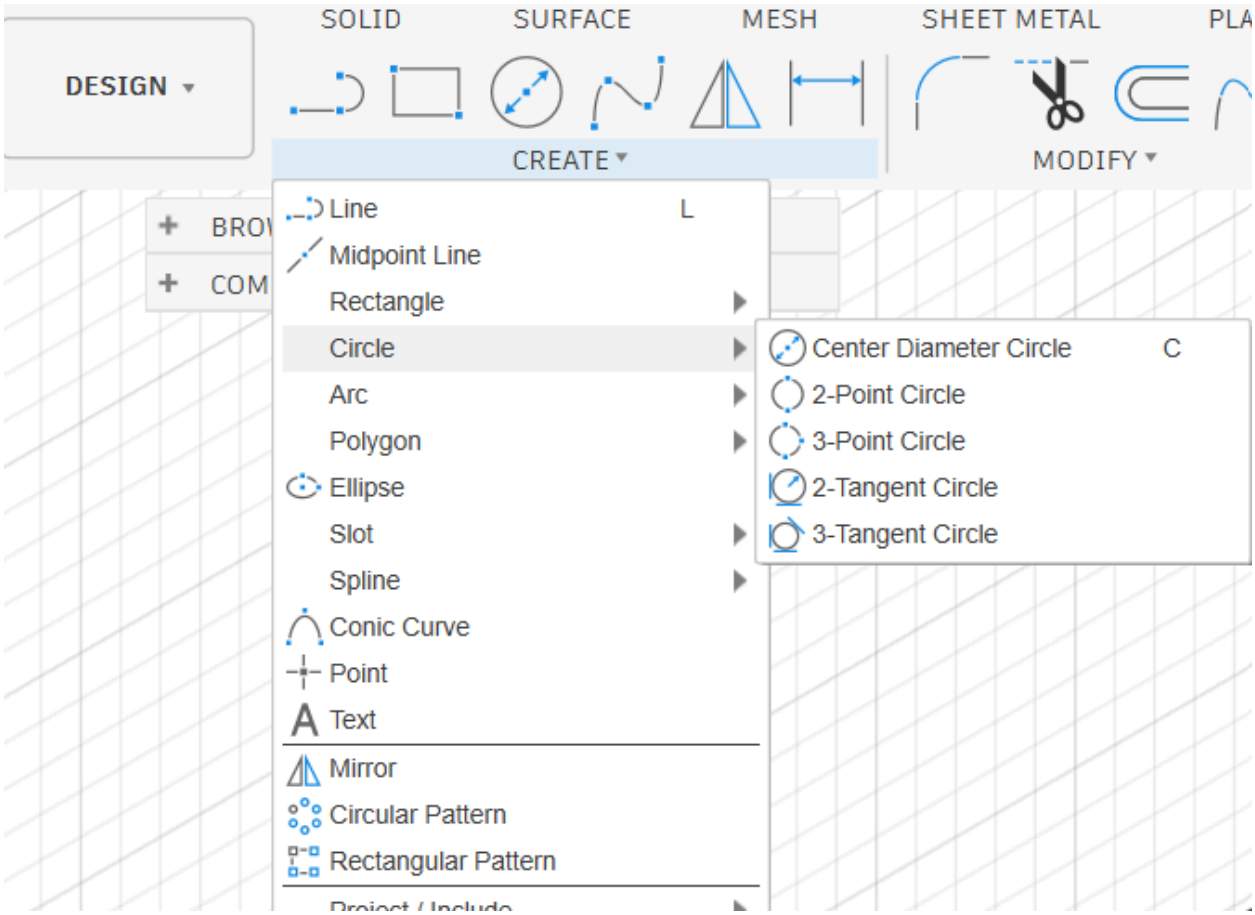
Choose Sketch > Polygon > Circumscribed Polygon and position the cursor at the origin of the sketch. Click to begin creating the polygon. Enter the specified dimensions as 40 mm and 30 mm, and choose 3 Side as given in the original procedure. Click Finish Sketch to complete the profile.

STEP-12: Extrude PENTAGON

Select Extrude and choose the polygon sketch profile. Set the extrusion distance to 10 mm and press ENTER to confirm.

STEP-13: Create a CIRCLE

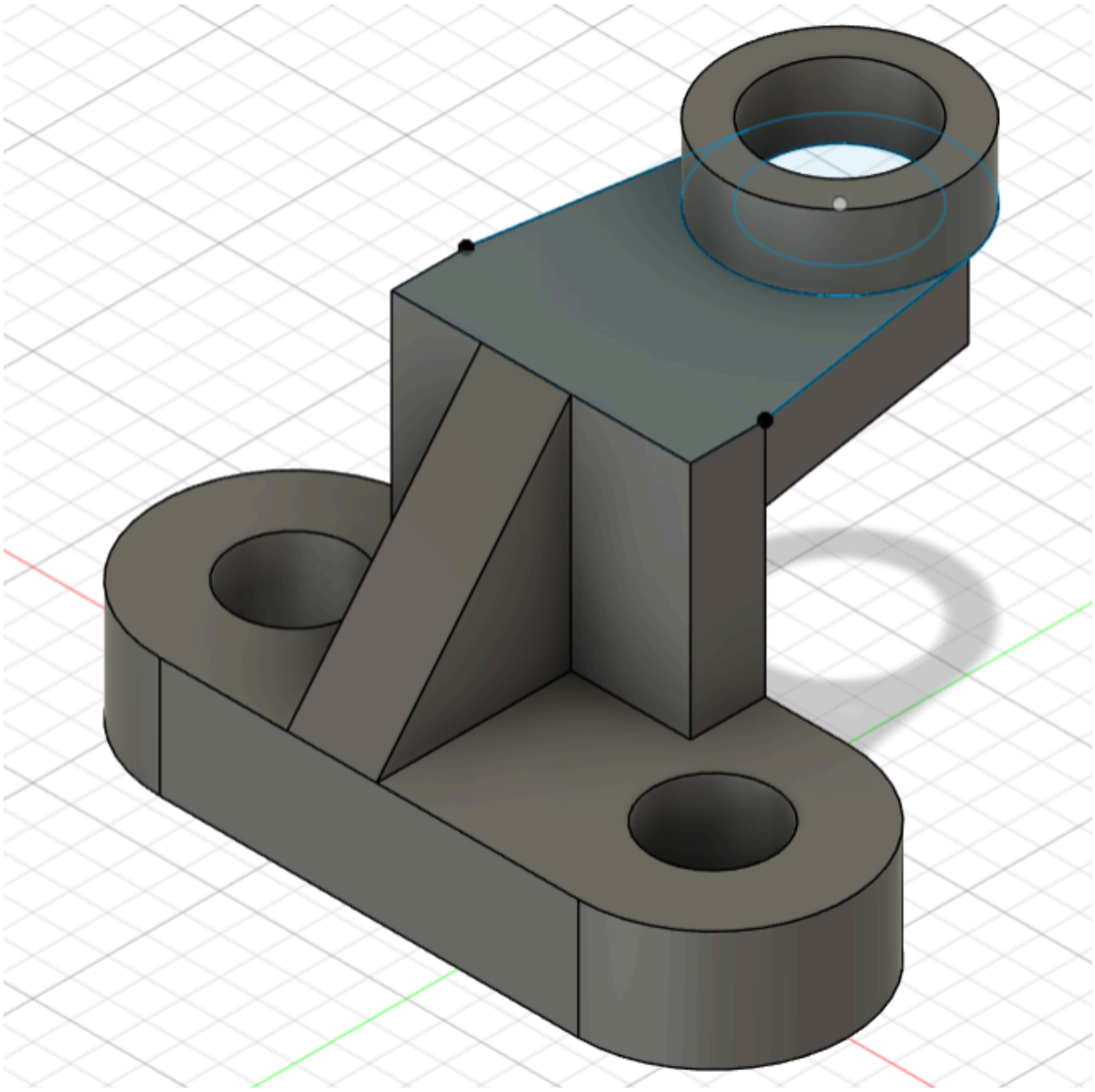
Select Sketch > Circle > Center Diameter Circle on the required end face of the polygon.



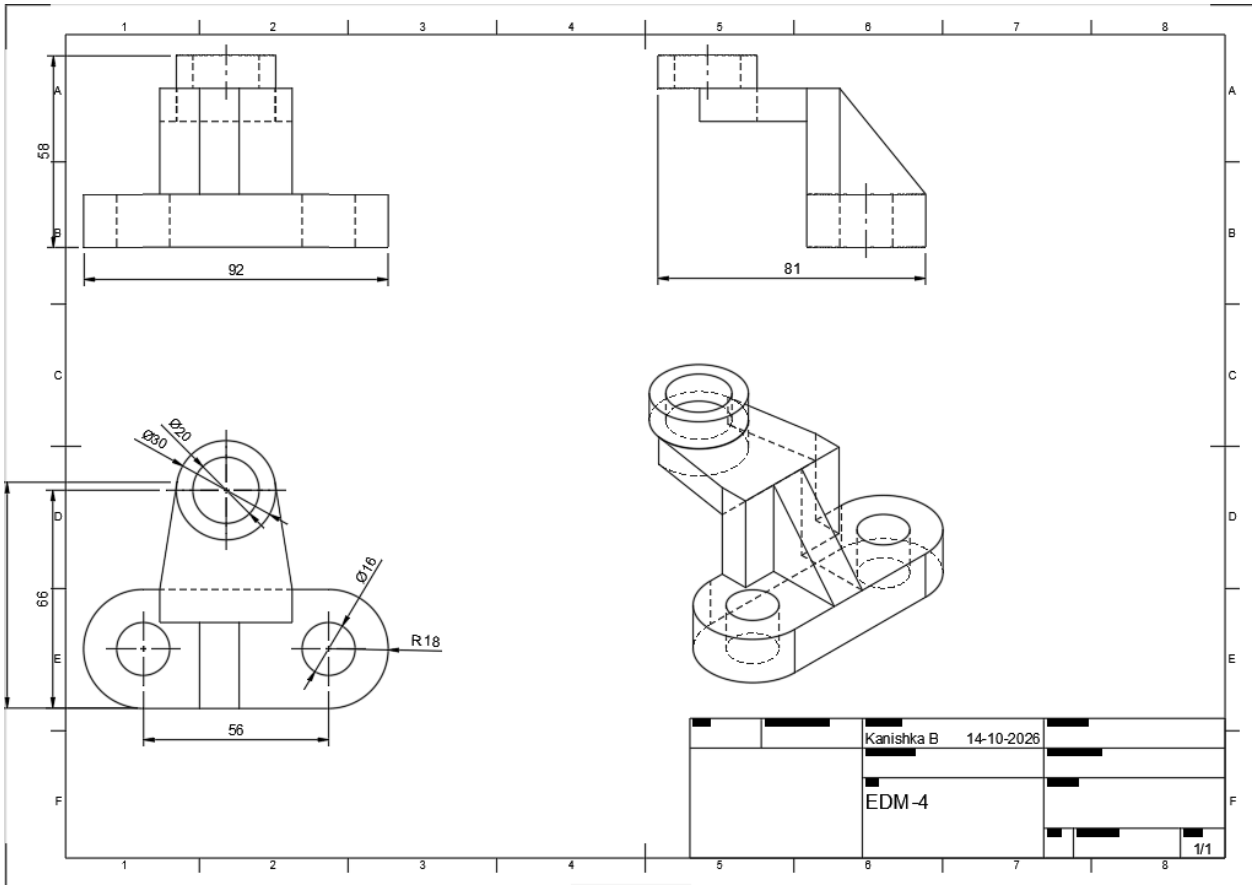
STEP-14: Extrude CIRCLE

Choose the Extrude command and select the required circular sketch profile. Set the extrusion distance to 10 mm on both sides and press ENTER to complete the operation.

INPUT:



Output:



RESULT:

Thus, the 3D model is created from the given isometric drawing using Fusion 360 tool.